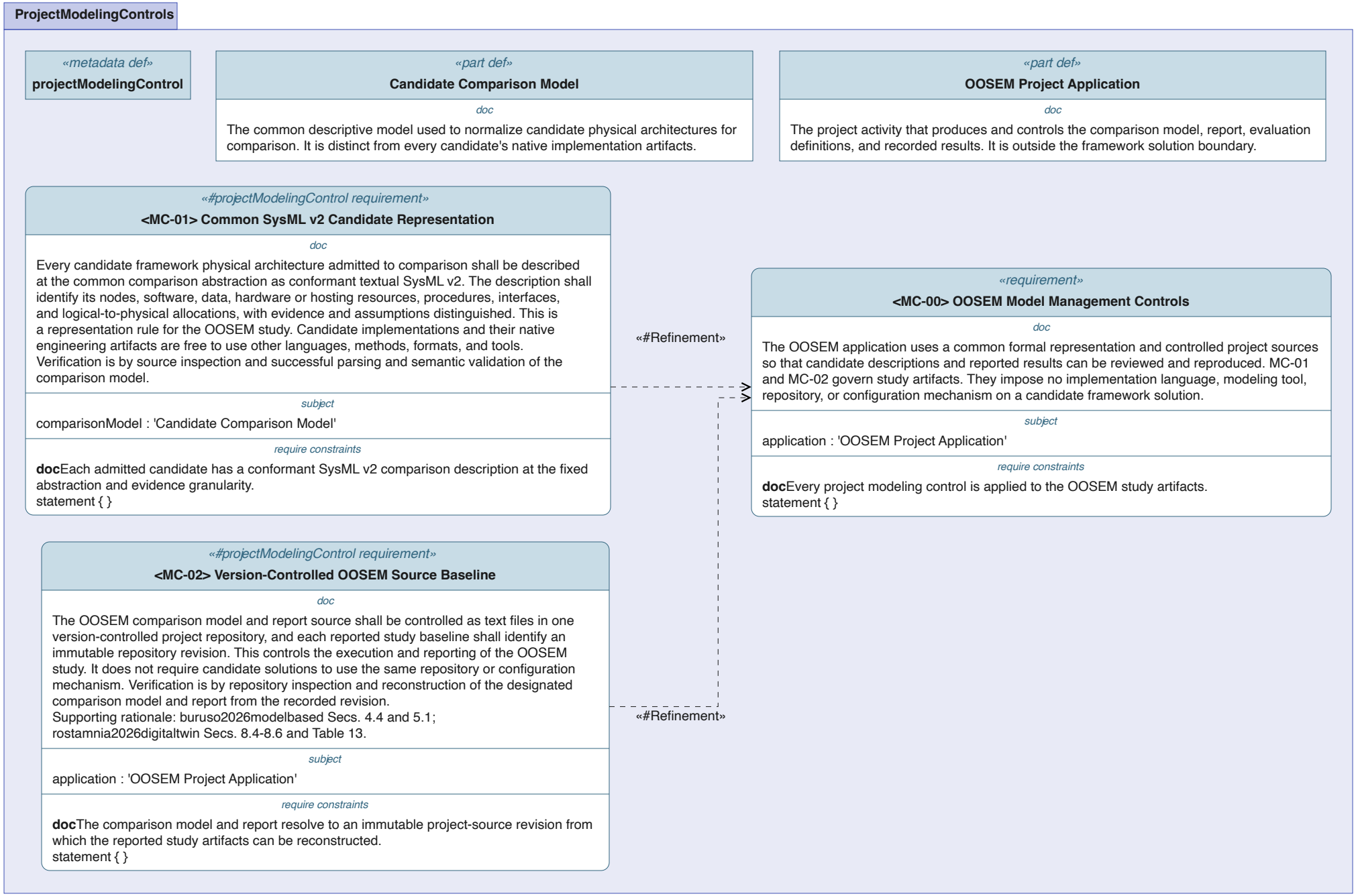
ProjectModelingControls
«metadata def»
projectModelingControl
«part def»
Candidate Comparison Model
doc
The common descriptive model used to normalize candidate physical architectures for comparison. It is distinct from every candidate's native implementation artifacts.
«part def»
OOSEM Project Application
doc
The project activity that produces and controls the comparison model, report, evaluation definitions, and recorded results. It is outside the framework solution boundary.
«#projectModelingControl requirement»
<MC-01> Common SysML v2 Candidate Representation
doc
Every candidate framework physical architecture admitted to comparison shall be described at the common comparison abstraction as conformant textual SysML v2. The description shall identify its nodes, software, data, hardware or hosting resources, procedures, interfaces, and logical-to-physical allocations, with evidence and assumptions distinguished. This is a representation rule for the OOSEM study. Candidate implementations and their native engineering artifacts are free to use other languages, methods, formats, and tools. Verification is by source inspection and successful parsing and semantic validation of the comparison model.
subject
comparisonModel : 'Candidate Comparison Model'
require constraints
docEach admitted candidate has a conformant SysML v2 comparison description at the fixed abstraction and evidence granularity.
statement { }
«#Refinement»
«requirement»
<MC-00> OOSEM Model Management Controls
doc
The OOSEM application uses a common formal representation and controlled project sources so that candidate descriptions and reported results can be reviewed and reproduced. MC-01 and MC-02 govern study artifacts. They impose no implementation language, modeling tool, repository, or configuration mechanism on a candidate framework solution.
subject
application : 'OOSEM Project Application'
require constraints
docEvery project modeling control is applied to the OOSEM study artifacts.
statement { }
«#projectModelingControl requirement»
<MC-02> Version-Controlled OOSEM Source Baseline
doc
The OOSEM comparison model and report source shall be controlled as text files in one version-controlled project repository, and each reported study baseline shall identify an immutable repository revision. This controls the execution and reporting of the OOSEM study. It does not require candidate solutions to use the same repository or configuration mechanism. Verification is by repository inspection and reconstruction of the designated comparison model and report from the recorded revision.
Supporting rationale: buruso2026modelbased Secs. 4.4 and 5.1; rostamnia2026digitaltwin Secs. 8.4-8.6 and Table 13.
subject
application : 'OOSEM Project Application'
require constraints
docThe comparison model and report resolve to an immutable project-source revision from which the reported study artifacts can be reconstructed.
statement { }
«#Refinement»